AI-BASED SMART VEHICLE HEALTH MONITORING SYSTEM

A Capstone Engineering Project Integrating Theoretical Computer Science, Soft Computing, and AIML

Department of Computer Engineering
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ABSTRACT

Modern connected automotive platforms generate continuous multi-sensor telemetry encompassing thermal, hydraulic, electrical, and mechanical vibrations. Traditional on-board diagnostic (OBD-II) units rely primarily on binary static threshold alerts that fail to address sensor noise, gradual degradation, or complex fault interdependencies. This capstone project introduces an end-to-end intelligent vehicle health monitoring platform founded on three pillars of modern computing: **(1) Theoretical Computer Science (TCS)** providing deterministic finite automata (DFA) state machines and formal SAE J2012 regular expression / context-free grammar parsers; **(2) Soft Computing (SC)** deploying a Mamdani Fuzzy Inference System (FIS) to quantify an un-crisp Vehicle Health Index (VHI) alongside genetic algorithm feature selection and neural learning; and **(3) Artificial Intelligence & Machine Learning (AIML)** utilizing probabilistic fault classifiers, Remaining Useful Life (RUL) regression, and multivariate anomaly detection. An interactive, zero-dependency web dashboard provides real-time telemetry simulation, ECU actuator governance, and diagnostic transparency.

1. CURRICULUM ALIGNMENT & SYSTEM ARCHITECTURE

To satisfy the academic objectives across Semester V subjects, the application divides the vehicle health lifecycle into three distinct theoretical layers:

Subject	Syllabus Modules Covered	Implemented Project Functionality
Theoretical Computer Science (TCS) Course: 24CSPCC501C	• Module I: Finite Automata, Moore/Mealy Machines • Module II: Regular Expressions & Languages • Module III & IV: CFG & Syntactic Parsers	• 5-state DFA / Moore Machine governing ECU limp mode and throttle. • Formal regex validator for SAE J2012 OBD-II DTCs. • Context-Free Grammar (CFG) parser for serialized CAN-bus telemetry.
Soft Computing (SC) Course: 24CSPCC502C	• Module 2: Fuzzy Sets, FIS, Centroid Defuzzification • Module 3: Multi-Layer Perceptron (MLP) • Module 4: Genetic Algorithms (GA) • Module 5: Hybrid Neuro-Fuzzy (ANFIS)	• 12-rule Mamdani Fuzzy System computing continuous Vehicle Health Index (0-100%). • Pure-NumPy MLP with backpropagation. • Binary Genetic Algorithm for telemetry feature subset optimization.

AI & Machine Learning (AIML)	• Supervised Multi-class Classification • Predictive Maintenance & Degradation Regression • Unsupervised Anomaly Detection	• Multi-class failure mode classifier (Cooling, Lubrication, Electrical, Drivetrain). • Physical wear Remaining Useful Life (RUL) estimation. • Multivariate Z-score anomaly detector.

2. THEORETICAL COMPUTER SCIENCE (TCS) SPECIFICATIONS

2.1 Formal DFA & Moore Machine Definition

The vehicle operating hierarchy is formally defined as a 6-tuple Moore Machine $M = (Q, \Sigma, \Delta, \delta, \lambda, q_0)$ where:
• **States Q:** {S■: NORMAL, S■: ADVISORY, S■: LIMP_MODE, S■: CRITICAL_FAULT, S■: EMERGENCY_HALT}
• **Input Alphabet Σ:** {ALL_NOMINAL, TRANSIENT_ANOMALY, PERSISTENT_MODERATE, SEVERE_THRESHOLD_BREACH, SAFETY_LOCKOUT, MANUAL_OVERRIDE_RESET}
• **Output Alphabet Δ:** Vector of {Throttle_Cap_%, MIL_Lamp, Audio_Buzzer, Fuel_Cutoff}
• **Output Mapping λ(q):** S■ → (100%, OFF, OFF, 0); S■ → (45%, FLASH_AMBER, INTERMITTENT, 0); S■ → (0%, SOLID_RED, CONTINUOUS, 1).
This guarantees that the vehicle ECU executes safety interventions deterministically without unbounded race conditions.

2.2 OBD-II Diagnostic Trouble Code (DTC) Regular Language

Per SAE J2012, all diagnostic trouble codes form a regular language recognized by DFA with regular expression:
L(DTC) = ^([PCBU])([0-3])([0-9A-F])([0-9A-F]{2})$
where character 1 identifies the domain (P=Powertrain, C=Chassis, B=Body, U=Network), character 2 specifies standards compliance (0=SAE standardized, 1=Manufacturer proprietary), and character 3 indexes the automotive subsystem.

3. SOFT COMPUTING (SC) FUZZY INFERENCE MODELING

Because sensor fluctuations occur gradually under real road vibrations and environmental ambient changes, crisp boolean thresholds cause false alarms. The Mamdani Fuzzy Inference System models four continuous input variables using triangular (trimf) and trapezoidal (trapmf) membership functions:
• **Engine Temperature:** Cool (-20..80°C), Normal (75..102°C), High (98..120°C), Overheating (115..150°C)
• **Oil Pressure:** Critical_Low (0..25 PSI), Low (20..36 PSI), Normal (32..68 PSI), High (62..100 PSI)
• **Defuzzification (Centroid Method / COG):** The aggregated fuzzy membership function μ(z) across the universe of discourse Z = [0, 100]% is calculated via:
$$VHI = (\int z \cdot \mu(z)\, dz) / (\int \mu(z)\, dz)$$
This continuous metric directly quantifies degradation severity prior to catastrophic mechanical seizure.

4. ARTIFICIAL INTELLIGENCE & MACHINE LEARNING (AIML)

The AIML layer delivers diagnostic classifications and physical prognostics:
• **Subsystem Fault Classifier:** Employs distance-weighted centroids with Softmax temperature scaling to classify failure modes into five categories: Normal Operation, Cooling System Overheat, Oil Starvation, Alternator Degradation, and Transmission Slip.
• **Predictive Maintenance RUL:** Estimates Remaining Useful Life (RUL in kilometers) by penalizing standard service intervals using non-linear thermal, lubrication, and vibrational stress coefficients:
$$RUL = (RUL_0 \cdot [VHI / 100]) / [1.0 + 2.5 \cdot Stress_thermal + 3.5 \cdot Stress_lube + 1.8 \cdot Stress_vib]$$
• **Multivariate Anomaly Detection:** Tracks Mahalanobis-like statistical distance from healthy vehicle telemetry baselines

to flag unknown zero-day mechanical anomalies.

5. EXPERIMENTAL RESULTS & SCENARIO EVALUATION

Scenario	Key Sensor Deviation	TCS State	Fuzzy VHI	AIML Classification	Est. RUL
Normal Highway Cruise	Nominal (90°C, 48 PSI, 14.1V)	S0_NORMAL	91.6% (Pristine)	NORMAL_OPERATION (64%)	13,732 km
Coolant Leak / Boil	Temp 123.5°C, DTC P0217	S3_CRITICAL	12.2% (Hazard)	COOLING_OVERHEAT (44%)	417 km
Oil Pump Failure	Oil 10.1 PSI, DTC P0524	S3_CRITICAL	23.0% (Hazard)	OIL_STARVATION (54%)	415 km
Alternator Failure	Voltage 10.29V, DTC P0562	S3_CRITICAL	43.3% (Degraded)	ELECTRICAL_DECAY (68%)	6,498 km
Transmission Shudder	Vib 8.6 m/s², RPM 3924	S3_CRITICAL	43.3% (Degraded)	TRANSMISSION_SLIP (66%)	1,051 km

6. VIVA VOCE & FACULTY DEFENSE CHEATSHEET

Q1: Why use both DFA (TCS) and Fuzzy Logic (SC) instead of just one?
Answer: Fuzzy Logic excels at handling continuous uncertainty and noise in sensor data to produce a gradient health score (0-100%). However, automotive safety actuators (throttle limits, fuel cutoff) cannot be fuzzy—they must execute deterministically. The DFA acts as the formal supervisory safety governor that translates fuzzy risk levels into provably safe operational states.

Q2: How does the SAE J2012 regular expression fit into theoretical computer science?
Answer: The diagnostic trouble code format represents a regular language *L(DTC)* over the ASCII alphabet. Because it can be recognized by a 5-state Deterministic Finite Automaton without memory/stack, it occupies the lowest level (Type-3) of the Chomsky hierarchy.

Q3: What defuzzification method is used and why?
Answer: The Centroid (Center of Gravity) method is used because it considers the entire shape of the aggregated fuzzy set, providing smooth, continuous output transitions as vehicle sensor readings fluctuate.